Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 101503 | *Subject: Geotechnical Engineering-I* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin
*(ii) There are **NINE** questions in this paper*
*(iii) Attempt **FIVE** questions in all*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Specific gravity refers to the ratio of the weight of soil solids to:
(i) Weight of equal volume of water
(ii) Unit weight of wet soil
(iii) Weight of dry air
(iv) Volume of air voids

(b) If the moisture content of a soil is 20% and the dry weight is 100g, what is the weight of water?
(i) 10g
(ii) 20g
(iii) 25g
(iv) 30g

(c) Plasticity of soil refers to the:
(i) Soil's ability to retain moisture
(ii) Soil's ability to undergo deformation without cracking
(iii) Soil's resistance to deformation
(iv) Soil's permeability

(d) A soil sample has a shrinkage limit of 10% and specific gravity of soil solids of 2.7. The porosity of the soil at shrinkage limit is:
(i) 21.2%
(ii) 52.7%
(iii) 30%
(iv) 70%

(e) Which instrument is commonly used to determine liquid limit?
(i) Proctor apparatus
(ii) CBR apparatus
(iii) Casagrande apparatus
(iv) Core cutter

(f) Darcy's law states that the velocity of flow is directly proportional to:
(i) Area
(ii) Hydraulic gradient
(iii) Time
(iv) Soil weight

(g) Maximum dry density is achieved at:
(i) Zero moisture
(ii) Optimum moisture
(iii) Air-dried condition
(iv) High moisture

(h) Contact pressure under a rigid footing is generally:
(i) Parabolic
(ii) Uneven
(iii) Sinusoidal
(iv) Uniform

(i) The ratio of volume of water present in a given soil mass to the total volume of its void is known as
(i) Porosity
(ii) Percentage voids
(iii) Void ratio
(iv) Degree of saturation

(j) Permeability is highest in:
(i) Clay
(ii) Silt
(iii) Silty clay
(iv) Gravel

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Q.2 (a) What is meant by a three-phase soil system? Draw a neat phase diagram showing weight and volume relationships. [7]

(b) A saturated clayey soil sample was tested in the laboratory. It has a specific gravity (G) of 2.70 and a water content (w) of 25%. determine the: [7]
(i) Void ratio (e)
(ii) Degree of saturation (S)

Q.3 (a) Explain the significance of the liquid limit, plastic limit, and shrinkage limit in soil mechanics. Sketch the Atterberg limits diagram. [7]

(b) Two soil samples A and B have the following properties: [7]
(i) Soil A: LL = 60%, PL = 40%, Clay = 35%
(ii) Soil B: LL = 60%, PL = 30%, Clay = 50%
Compare their plasticity indices and activities, and identify which is more problematic in terms of swelling/shrinkage.

Q.4 (a) Compare the constant head and falling head permeability tests in terms of soil suitability and test setup. [7]

(b) Water flows through a soil sample of length 20 cm and cross-sectional area 50 cm^2. The head loss measured is 10 cm over 5 minutes and the discharge is 300 cm^3. Calculate the coefficient of permeability. [7]

Q.5 (a) State Darcy's Law. What are its assumptions and limitations? Under what soil conditions is Darcy's law not valid? [7]

(b) A soil profile consists of a 6 m thick homogeneous soil layer with the water table located at a depth of 3 m from the ground surface. The unit weight of soil above the water table is 18kN/m^3 and the saturated unit weight below the water table is 20.5kN/m^3. Assume γ_w =9.81kN/m^3. [7]
Determine the effective stress at a depth of 6 m below the ground surface. Illustrate your solution with a stress distribution diagram, showing total stress, pore water pressure, and effective stress with depth.

Q.6 (a) Differentiate between the Standard Proctor Test and the Modified Proctor Test for soil compaction. Explain their objectives and test procedures. [7]

(b) In a Standard Proctor test, a maximum dry density of 1.85 g/cm^3 is obtained at 12% water content. What is the dry density at 14% moisture content if bulk density is 2.00 g/cm^3? [7]

Q.7 (a) State Boussinesq's assumptions for stress distribution in a semi-infinite soil mass. When is this theory not valid? [7]

(b) Differentiate between contact pressure under rigid and flexible footings. How does the distribution vary? [7]

Q.8 (a) State the assumptions of Terzaghi's 1D consolidation theory. Discuss its limitations in real field conditions. [7]

(b) Derive the relationship between: [7]
(i)Void-ratio and porosity
(ii) Moisture content and degree of saturation

Q.9 (a) Differentiate between compaction, normally consolidated and over consolidated soils. How would you determine the over consolidation pressure? [7]

(b) A concrete dam is constructed across a river over a permeable stratum of soil of limited thickness. The water heads are upstream side 16m and 2m on the downstream side. The flow net constructed under the dam gives N_f=7 and N_d= 12. Calculate the seepage loss through the subsoil if the average value of the hydraulic conductivity is $6x10^{-3}$ cm/sec horizontally and $3x10^{-4}$ cm/sec vertically. Calculate the exit gradient if the average length of the last field is 0.9 m. [7]

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